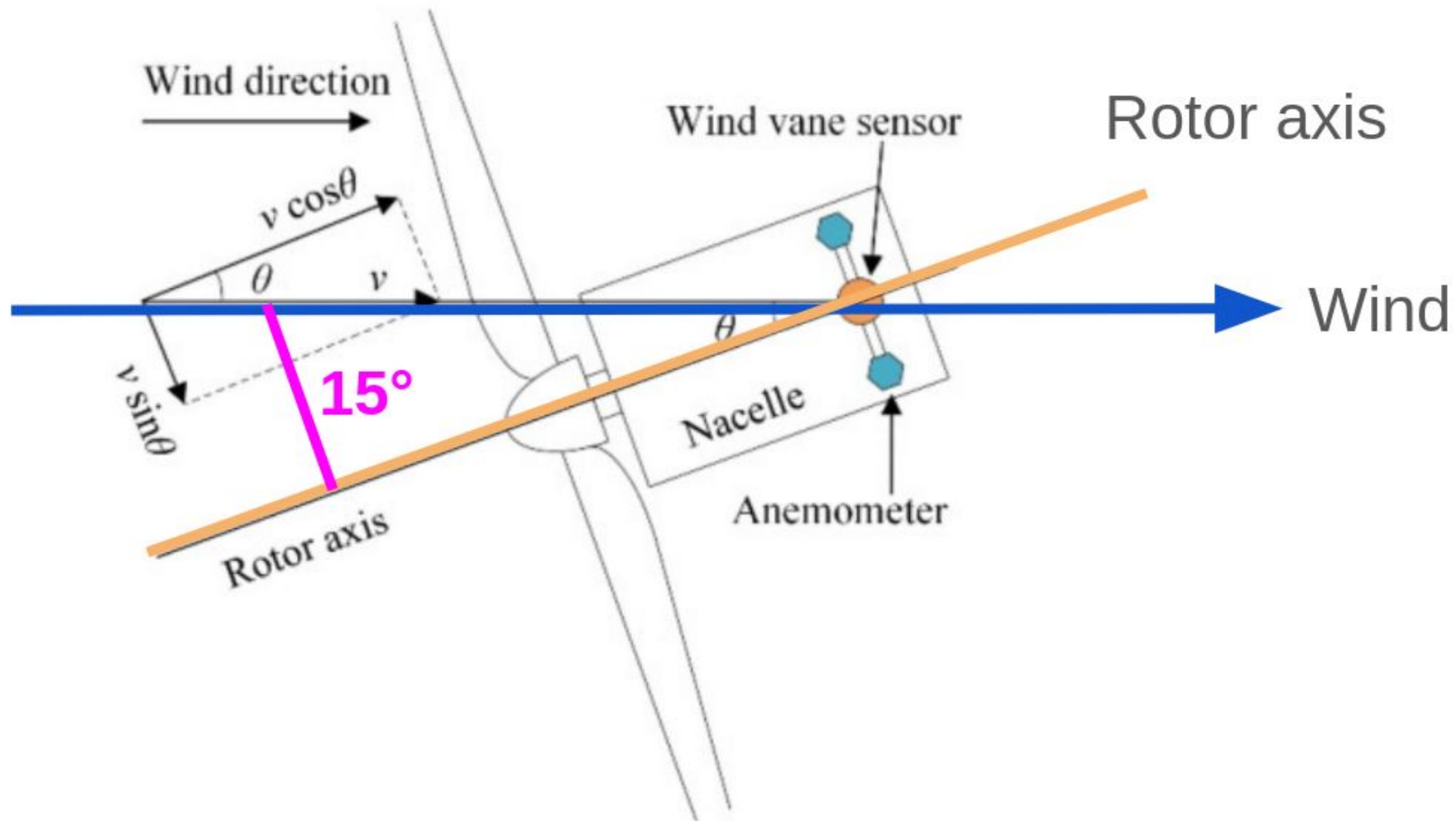


WeDoWind

How to determine the yaw-misalignment in a cost effective manner for NADARA

Energy Data Hackdays • 10.09.2026





A 15° misalignment can cost $\approx 10\%$ of power

What happens?

Wind must be aligned with the rotor to maximize energy production.

Today's approach

Add an extra sensor to measure yaw alignment.

Why not?

$\approx$ CHF 200k per turbine $\rightarrow \approx$ CHF 20M for 100 turbines.

So we asked: can we infer the missing signal from SCADA data we already have?

Use the fleet to estimate the wind direction



Key idea: constant error terms cancel out when we compare turbines.

The data contains the signal we need

Participant 33

T0 · no labels

Sub #0

Band 2

1.06

RMSE (°)

0.94

MAE (°)

Participant 33

T3 · 3 turbines

Sub #1

Band 2

1.58

RMSE (°)

needs 0.90 to beat band lead

1.49

MAE (°)

Result

Yaw misalignment can be estimated without installing an additional sensor.

Why it matters

A scalable, data-driven alternative for wind parks.

THE TAKEAWAY

**Don't add hardware
before asking what the data already knows.**

WeDoWind • From existing SCADA data → hidden yaw insight

Thank you